

MSX Emulator User Guide

MSX MODELS: STARTUP OPTIONS AND COMMAND
CONSOLE REFERENCE

IGNACIO CEA FORNIÉS

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IGNACIO CEA FORNIÉS

Foreword

This guide describes the startup and command interfaces implemented by the two MSX executables in EMULATORS: MSXEmulator (without a local command console) and MSXEmulatorC (with a local command console). It covers the implemented MSX model selectors, PAL and NTSC operation, model-dependent ROM languages, and is based on both entry points, the MSX::MSXEmulator inheritance chain, LocalConsole, and the effective MSX::CommandBuilder -> MCEmul::StandardCommandBuilder -> MCEmul::StandardCommandBuilder chain constructed by the entry points.

Command and parameter names are shown exactly as accepted by the current code. Startup options use /XVALUE with no space between the one-letter option and its value. Command names are matched as written by the builders; use uppercase names for reliable operation. Positional command parameters are separated by spaces; examples use paths without embedded spaces.

The non-console executable does not expose an interactive prompt. When started with /o it creates a remote communication system using the MSX and standard command builders. Commands marked Local console only are intercepted by LocalConsole and therefore cannot be sent to the non-console executable.

1. Startup Parameters: Version Without a Local Console

Executable: MSXEmulator.exe. It accepts the framework and MSX-specific options below, plus /o and the communication-port option. Option letters and language identifiers are case-sensitive because the command-line parser preserves them exactly.

/o

Meaning: Enables or disables the TCP remote-command system in the executable without a local console.

General form:

```
/o[YES|NO]
```

- YES or an empty value: create and start the remote command system.
- NO (and any other non-YES non-empty value): leave it disabled.
- Omitted: no remote listener is created.

Examples:

```
MSXEmulator.exe /o  
MSXEmulator.exe /oYES /p61000  
MSXEmulator.exe /oNO
```

Consequence: When enabled, remote clients can execute the MSX and standard builder commands. LocalConsole-only commands remain unavailable.

/p

Meaning: Selects the TCP listening port used by the non-console executable's remote command channel.

General form:

```
/pPORTNUMBER
```

- 0 through 65535: requested listening port; values above 1000 are recommended.
- Omitted: port 60000 is used when /o enables the channel.
- Collision: the inherited emulator also interprets /p as a comma-separated peripheral-ID list. A supplied port number is therefore also attempted as a peripheral ID during initialization.

Examples:

```
MSXEmulator.exe /o  
MSXEmulator.exe /o /p61000
```

Consequence: The remote server listens on the selected port. When /p is present, the same value is also attempted as a peripheral ID and normally produces a log warning.

/h

Meaning: Prints the executable help and prevents the emulation from starting.

General form:

```
/h
```

- No value.

Examples:

```
MSXEmulator.exe /h
```

Consequence: Initialization returns failure after printing the help; the current main program therefore exits with code 1.

/f

Meaning: Loads a binary file whose header contains the destination address.

General form:

```
/fFILE
```

- FILE: path to the header-addressed binary file; attach it directly to /f.

Examples:

```
MSXEmulator.exe /fprogram.bin  
MSXEmulator.exe /fgames\demo.bin /a$080d
```

Consequence: The file is loaded before execution. An unreadable or invalid file aborts initialization.

/k

Meaning: Loads a block-structured memory file.

General form:

```
/kFILE
```

- FILE: path to a block file understood by the framework.

Examples:

```
MSXEmulator.exe /ksnapshot.blocks
```

Consequence: Every encoded block is placed at its stored address. A load error aborts initialization.

/c

Meaning: Parses, assembles and loads an assembler source file.

General form:

```
/cFILE
```

- FILE: assembler source path.

Examples:

```
MSXEmulator.exe /cstartup.asm /s
```


Consequence: The assembled bytes are installed before the main loop. Syntax or load errors are logged and initialization stops.

/l

Meaning: Selects the framework diagnostic level.

General form:

```
/lLEVEL
```

- 0: no diagnostics.
- 1: errors (default).
- 2: dump at exit.
- 3: errors and warnings.
- 4: all regular diagnostics.
- 5: internal trace diagnostics.

Examples:

```
MSXEmulator.exe /l0  
MSXEmulator.exe /l4
```

Consequence: Higher values produce progressively more diagnostic output and may reduce performance.

/a

Meaning: Overrides the address at which execution begins.

General form:

```
/aADDRESS
```

- ADDRESS: MSX address in the framework number syntax, for example decimal or hexadecimal with \$.

Examples:

```
MSXEmulator.exe /a$080d  
MSXEmulator.exe /fprogram.bin /a2061
```

Consequence: A non-zero, valid address replaces the normal reset/program entry address.

/d

Meaning: Creates startup stop points at one or more addresses.

General form:

```
/dADDRESS1,ADDRESS2,...
```

- ADDRESSn: comma-separated address. Each entry accepts the framework number syntax.

Examples:

```
MSXEmulator.exe /d$080d  
MSXEmulator.exe /d$080d,$ffd2 /s
```

Consequence: Execution stops when the CPU reaches any listed address, allowing later inspection or continuation.

/s

Meaning: Controls whether the emulated computer starts stopped.

General form:

```
/s[YES|NO]
```

- YES or an empty value: start stopped.
- NO (and any non-YES non-empty value): start running.
- Omitted: start running.

Examples:

```
MSXEmulator.exe /s  
MSXEmulator.exe /sNO
```

Consequence: When enabled, the machine is initialized but the CPU does not run until a RUN/NEXT command or the equivalent UI action is issued.

/i

Meaning: Selects the ROM/keyboard language used by the selected MSX model.

General form:

```
/iLANG
```

- SVI728: ENG (default) or ESP; other values make ROM loading fail.
- SONYHB10P: always uses its English ROM; the supplied value is ignored.
- PHILIPSVG8010: FRA selects French; ENG and unrecognized values use English.
- CANONV20: always uses its English ROM; the supplied value is ignored.
- SVI738: the code accepts ENG, DEU and SEW. The help banner says SWE, but that spelling is not accepted; SVI738 is currently incomplete and must not be selected for normal use.
- MSXSTD/default: always uses the English SVI728-compatible ROM.
- Language identifiers are case-sensitive; use the uppercase forms above.

Examples:

```
MSXEmulator.exe /mSVI728 /iESP  
MSXEmulator.exe /mPHILIPSVG8010 /iFRA
```

Consequence: The selected model loads its matching BASIC/BIOS ROM before initialization. The required file must exist under ./bios relative to the process working directory.

/r

Meaning: Selects the initial AY-3-8910 PSG output state.

General form:

```
/r[YES|NO]
```

- YES or an empty value: sound enabled.
- NO (and any other non-YES non-empty value): sound disabled.
- Omitted: sound enabled.

Examples:

```
MSXEmulator.exe /r  
MSXEmulator.exe /rNO
```

Consequence: The active model's PSG starts enabled or disabled as requested.

/g

Meaning: Controls the CRT post-processing effect at startup.

General form:

```
/g[YES|NO]
```

- YES or an empty value: CRT effect enabled.
- NO (and any non-YES non-empty value): disabled.
- Omitted: disabled.

Examples:

```
MSXEmulator.exe /g  
MSXEmulator.exe /gNO
```

Consequence: The screen starts with or without CRT emulation. CRTON and CRTOFF can change it later.

/n

Meaning: Requests NTSC timing and the TMS9918A instead of PAL timing and the TMS9929A.

General form:

```
/n
```

- No value: presence selects NTSC for each recognized named model; omission selects PAL.
- The default MSXSTD model is fixed to PAL, so /n has no effect when /m is omitted or unknown.

Examples:

```
MSXEmulator.exe /mSVI728 /n  
MSXEmulator.exe /mCANONV20
```

Consequence: A recognized named model uses the requested PAL/NTSC VDP and geometry. The MSX CPU clock remains 3.58 MHz in the implemented MSX1 models.

/w

Meaning: Requests a model memory configuration.

General form:

```
/wCONF
```

- 0 or omitted: the only configuration currently admitted by every defined model.
- Other values: rejected by the selected model and adjusted back to configuration 0.

- Configuration 0 provides 16 KiB RAM for MSXSTD, SVI728, SONYHB10P and the incomplete SVI738; 32 KiB for PHILIPSVG8010; and 64 KiB for CANONV20.

Examples:

```
MSXEmulator.exe /mSVI728 /w0
MSXEmulator.exe /mCANONV20 /w0
```

Consequence: The model validates the request and uses its only basic mapping. /w does not currently change RAM capacity; /m selects the model and therefore the installed RAM.

/b

Meaning: Draws the diagnostic grid over the writable screen area and selects its startup colour.

General form:

```
/b[COLOR]
```

- COLOR omitted or 0: colour 0.
- 1 through 3: requested startup grid colour.
- Any value greater than 3: clamped to 3 by the startup parser.

Examples:

```
MSXEmulator.exe /b
MSXEmulator.exe /b3
```

Consequence: The MSX screen starts with its diagnostic grid enabled in the selected colour.

/m

Meaning: Selects the MSX machine model and its ROM, memory map and hardware profile.

General form:

```
/mMACHINE
```

- SVI728: Spectravideo SVI-728 MSX1, 16 KiB RAM; English or Spanish ROM.
- SONYHB10P: Sony HB-10P MSX1, 16 KiB RAM; English ROM.
- PHILIPSVG8010: Philips VG-8010 MSX1, 32 KiB RAM; English or French ROM.
- CANONV20: Canon V-20 MSX1, 64 KiB RAM; English ROM.
- SVI738: advertised as Spectravideo SVI-738/MSX2 with 16 KiB RAM, but its VDP and PSG factories return null and the implementation is not operational.
- Omitted or unknown: MSXSTD, a PAL-only standard MSX1 model equivalent to an English SVI728 with 16 KiB RAM. Unknown names are logged.

Examples:

```
MSXEmulator.exe /mSVI728 /iESP
MSXEmulator.exe /mCANONV20 /n
```

Consequence: The selected model constructs its corresponding memory, VDP, PSG and ROM profile. Do not select SVI738 until its missing devices are implemented.

2. Startup Parameters: Version With a Local Console

Executable: MSXEmulatorC.exe. Its main program delegates the complete command line to MSX::MSXEmulator, so all inherited framework and MSX-specific options are active even though the short main-level banner only advertises /h. It has no /o listener; commands are entered in its local console.

/h

Meaning: Prints the executable help and prevents the emulation from starting.

General form:

```
/h
```

- No value.

Examples:

```
MSXEmulatorC.exe /h
```

Consequence: Initialization returns failure after printing the help; the current main program therefore exits with code 1.

/f

Meaning: Loads a binary file whose header contains the destination address.

General form:

```
/fFILE
```

- FILE: path to the header-addressed binary file; attach it directly to /f.

Examples:

```
MSXEmulatorC.exe /fprogram.bin  
MSXEmulatorC.exe /fgames\demo.bin /a$080d
```

Consequence: The file is loaded before execution. An unreadable or invalid file aborts initialization.

/k

Meaning: Loads a block-structured memory file.

General form:

```
/kFILE
```

- FILE: path to a block file understood by the framework.

Examples:

```
MSXEmulatorC.exe /ksnapshot.blocks
```

Consequence: Every encoded block is placed at its stored address. A load error aborts initialization.

/c

Meaning: Parses, assembles and loads an assembler source file.

General form:

```
/cFILE
```

- FILE: assembler source path.

Examples:

```
MSXEmulatorC.exe /cstartup.asm /s
```

Consequence: The assembled bytes are installed before the main loop. Syntax or load errors are logged and initialization stops.

/l

Meaning: Selects the framework diagnostic level.

General form:

```
/lLEVEL
```

- 0: no diagnostics.
- 1: errors (default).
- 2: dump at exit.
- 3: errors and warnings.
- 4: all regular diagnostics.
- 5: internal trace diagnostics.

Examples:

```
MSXEmulatorC.exe /l0  
MSXEmulatorC.exe /l4
```

Consequence: Higher values produce progressively more diagnostic output and may reduce performance.

/a

Meaning: Overrides the address at which execution begins.

General form:

```
/aADDRESS
```

- ADDRESS: MSX address in the framework number syntax, for example decimal or hexadecimal with \$.

Examples:

```
MSXEmulatorC.exe /a$080d  
MSXEmulatorC.exe /fprogram.bin /a2061
```

Consequence: A non-zero, valid address replaces the normal reset/program entry address.

/d

Meaning: Creates startup stop points at one or more addresses.

General form:

```
/dADDRESS1,ADDRESS2,...
```

- ADDRESSn: comma-separated address. Each entry accepts the framework number syntax.

Examples:

```
MSXEmulatorC.exe /d$080d  
MSXEmulatorC.exe /d$080d,$ffd2 /s
```

Consequence: Execution stops when the CPU reaches any listed address, allowing later inspection or continuation.

/s

Meaning: Controls whether the emulated computer starts stopped.

General form:

```
/s[YES|NO]
```

- YES or an empty value: start stopped.
- NO (and any non-YES non-empty value): start running.
- Omitted: start running.

Examples:

```
MSXEmulatorC.exe /s  
MSXEmulatorC.exe /sNO
```

Consequence: When enabled, the machine is initialized but the CPU does not run until a RUN/NEXT command or the equivalent UI action is issued.

/i

Meaning: Selects the ROM/keyboard language used by the selected MSX model.

General form:

```
/iLANG
```

- SVI728: ENG (default) or ESP; other values make ROM loading fail.
- SONYHB10P: always uses its English ROM; the supplied value is ignored.
- PHILIPSVG8010: FRA selects French; ENG and unrecognized values use English.
- CANONV20: always uses its English ROM; the supplied value is ignored.
- SVI738: the code accepts ENG, DEU and SEW. The help banner says SWE, but that spelling is not accepted; SVI738 is currently incomplete and must not be selected for normal use.
- MSXSTD/default: always uses the English SVI728-compatible ROM.
- Language identifiers are case-sensitive; use the uppercase forms above.

Examples:

```
MSXEmulatorC.exe /mSVI728 /iESP
```

```
MSXEmulatorC.exe /mPHILIPSVG8010 /iFRA
```

Consequence: The selected model loads its matching BASIC/BIOS ROM before initialization. The required file must exist under ./bios relative to the process working directory.

/r

Meaning: Selects the initial AY-3-8910 PSG output state.

General form:

```
/r[YES|NO]
```

- YES or an empty value: sound enabled.
- NO (and any other non-YES non-empty value): sound disabled.
- Omitted: sound enabled.

Examples:

```
MSXEmulatorC.exe /r  
MSXEmulatorC.exe /rNO
```

Consequence: The active model's PSG starts enabled or disabled as requested.

/p

Meaning: Connects one or more peripherals during startup.

General form:

```
/pPER1[,PER2...]
```

- 10: typewriter keyboard injection.
- 101: MSX CAS cassette injection through standard BIOS traps.

Examples:

```
MSXEmulatorC.exe /p10  
MSXEmulatorC.exe /p10,101
```

Consequence: Each requested peripheral is built and connected to the first compatible device. Unsupported IDs are logged and initialization continues.

/g

Meaning: Controls the CRT post-processing effect at startup.

General form:

```
/g[YES|NO]
```

- YES or an empty value: CRT effect enabled.
- NO (and any non-YES non-empty value): disabled.
- Omitted: disabled.

Examples:

```
MSXEmulatorC.exe /g  
MSXEmulatorC.exe /gNO
```


Consequence: The screen starts with or without CRT emulation. CRTON and CROFF can change it later.

/n

Meaning: Requests NTSC timing and the TMS9918A instead of PAL timing and the TMS9929A.

General form:

```
/n
```

- No value: presence selects NTSC for each recognized named model; omission selects PAL.
- The default MSXSTD model is fixed to PAL, so /n has no effect when /m is omitted or unknown.

Examples:

```
MSXEmulatorC.exe /mSVI728 /n  
MSXEmulatorC.exe /mCANONV20
```

Consequence: A recognized named model uses the requested PAL/NTSC VDP and geometry. The MSX CPU clock remains 3.58 MHz in the implemented MSX1 models.

/w

Meaning: Requests a model memory configuration.

General form:

```
/wCONF
```

- 0 or omitted: the only configuration currently admitted by every defined model.
- Other values: rejected by the selected model and adjusted back to configuration 0.
- Configuration 0 provides 16 KiB RAM for MSXSTD, SVI728, SONYHB10P and the incomplete SVI738; 32 KiB for PHILIPSVG8010; and 64 KiB for CANONV20.

Examples:

```
MSXEmulatorC.exe /mSVI728 /w0  
MSXEmulatorC.exe /mCANONV20 /w0
```

Consequence: The model validates the request and uses its only basic mapping. /w does not currently change RAM capacity; /m selects the model and therefore the installed RAM.

/b

Meaning: Draws the diagnostic grid over the writable screen area and selects its startup colour.

General form:

```
/b[COLOR]
```

- COLOR omitted or 0: colour 0.
- 1 through 3: requested startup grid colour.
- Any value greater than 3: clamped to 3 by the startup parser.

Examples:

```
MSXEmulatorC.exe /b
```

```
MSXEmulatorC.exe /b3
```

Consequence: The MSX screen starts with its diagnostic grid enabled in the selected colour.

/m

Meaning: Selects the MSX machine model and its ROM, memory map and hardware profile.

General form:

```
/mMACHINE
```

- SVI728: Spectravideo SVI-728 MSX1, 16 KiB RAM; English or Spanish ROM.
- SONYHB10P: Sony HB-10P MSX1, 16 KiB RAM; English ROM.
- PHILIPSVG8010: Philips VG-8010 MSX1, 32 KiB RAM; English or French ROM.
- CANONV20: Canon V-20 MSX1, 64 KiB RAM; English ROM.
- SVI738: advertised as Spectravideo SVI-738/MSX2 with 16 KiB RAM, but its VDP and PSG factories return null and the implementation is not operational.
- Omitted or unknown: MSXSTD, a PAL-only standard MSX1 model equivalent to an English SVI728 with 16 KiB RAM. Unknown names are logged.

Examples:

```
MSXEmulatorC.exe /mSVI728 /iESP  
MSXEmulatorC.exe /mCANONV20 /n
```

Consequence: The selected model constructs its corresponding memory, VDP, PSG and ROM profile. Do not select SVI738 until its missing devices are implemented.

3. Command Console Reference

The local console checks its eleven emulator-level commands first. All remaining text is delegated through MSX::CommandBuilder and the standard builder chain. The current entry points pass a StandardCommandBuilder into MSX::CommandBuilder, whose constructor adds another StandardCommandBuilder parent; duplicate registrations resolve to the same standard command set. The remote /o channel in MSXEmulator.exe uses the same builder chain.

3.1 Standard Framework Commands

HELP

Availability: Local console and remote /o channel.

Accepted aliases: ? and CHELP.

Meaning: Displays the command help assembled for the MSX emulator.

General syntax:

```
HELP [COMMAND]
```

Parameters:

- COMMAND: optional command name whose detailed help should be shown; without it, the available help catalogue is displayed.

Examples:

```
HELP
HELP VDPMEMORY
? PSG
```

Result and consequences: Displays MSX-specific and standard builder help. In the local executable, LocalConsole also supplies help for its eleven emulator/media commands.

AUTHOR

Availability: Local console and remote /o channel.

Accepted aliases: CAUTHOR.

Meaning: Returns framework author information.

General syntax:

```
AUTHOR
```

Parameters:

- No parameters.

Examples:

```
AUTHOR
```

Result and consequences: Returns framework author information.

STATUS

Availability: Local console and remote /o channel.

Accepted aliases: CSTATUS.

Meaning: Shows the Z80 status register.

General syntax:

```
STATUS
```

Parameters:

- No parameters.

Examples:

```
STATUS
```

Result and consequences: Shows the Z80 status register.

PC

Availability: Local console and remote /o channel.

Accepted aliases: CPC.

Meaning: Shows the current program counter.

General syntax:

```
PC
```

Parameters:

- No parameters.

Examples:

```
PC
```

Result and consequences: Shows the current program counter.

REGISTERS

Availability: Local console and remote /o channel.

Accepted aliases: CREGISTERS.

Meaning: Shows all CPU registers.

General syntax:

```
REGISTERS
```

Parameters:

- No parameters.

Examples:

```
REGISTERS
```

Result and consequences: Shows all CPU registers.

CHANGereg

Availability: Local console and remote /o channel.

Accepted aliases: CCHANGereg.

Meaning: Changes one CPU internal register.

General syntax:

```
CHANGereg REGNAME VALUE
```

Parameters:

- REGNAME: registered Z80 register name.
- VALUE: new decimal, z-prefixed binary, o-prefixed octal or \$-prefixed hexadecimal value.

Examples:

```
CHANGereg A $41
```

Result and consequences: The register changes only if it exists and accepts the supplied byte width; otherwise an error is returned.

STACK

Availability: Local console and remote /o channel.

Accepted aliases: CSTACK.

Meaning: Shows stack state.

General syntax:

```
STACK [ALL]
```

Parameters:

- ALL: optional; includes stack contents in addition to basic state.

Examples:

```
STACK  
STACK ALL
```

Result and consequences: Shows stack state.

CPUSTATUS

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSTATUS.

Meaning: Shows registers, status, program counter and stack.

General syntax:

```
CPUSTATUS
```

Parameters:

- No parameters.

Examples:

```
CPUSTATUS
```

Result and consequences: Shows registers, status, program counter and stack.

CPUSSTATUS

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSSTATUS.

Meaning: Shows the compact CPU status without stack detail.

General syntax:

```
CPUSSTATUS
```

Parameters:

- No parameters.

Examples:

```
CPUSSTATUS
```

Result and consequences: Shows the compact CPU status without stack detail.

CPUSTATE

Availability: Local console and remote /o channel.

Accepted aliases: CCPUSTATE.

Meaning: Shows the CPU execution state and state-specific data.

General syntax:

```
CPUSTATE
```

Parameters:

- No parameters.

Examples:

```
CPUSTATE
```

Result and consequences: Shows the CPU execution state and state-specific data.

CPUINFO

Availability: Local console and remote /o channel.

Accepted aliases: CCPUINFO.

Meaning: Returns the complete CPU information structure.

General syntax:

```
CPUINFO
```

Parameters:

- No parameters.

Examples:

```
CPUINFO
```

Result and consequences: Returns the complete CPU information structure.

MEMORY

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORY.

Meaning: Reads the active CPU-visible memory range.

General syntax:

```
MEMORY INITIAL [FINAL]
```

Parameters:

- INITIAL: first address.
- FINAL: optional last address; with no final address only the initial location is reported.

Examples:

```
MEMORY $0400  
MEMORY $0400 $0427
```

Result and consequences: Reads the active CPU-visible memory range.

MEMORYDUMP

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYDUMP.

Meaning: Dumps every memory view between two addresses, including inactive RAM/ROM views.

General syntax:

```
MEMORYDUMP INITIAL FINAL
```

Parameters:

- INITIAL: first address.
- FINAL: last address.

Examples:

```
MEMORYDUMP $a000 $a03f
```

Result and consequences: Dumps every memory view between two addresses, including inactive RAM/ROM views.

MEMORYSTR

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYSTR.

Meaning: Shows the configured memory structure.

General syntax:

```
MEMORYSTR [ALL]
```

Parameters:

- ALL: optional; includes inactive regions. Without it only active regions are shown.

Examples:

```
MEMORYSTR  
MEMORYSTR ALL
```

Result and consequences: Shows the configured memory structure.

SETMEMORY

Availability: Local console and remote /o channel.

Accepted aliases: CSETMEMORY.

Meaning: Writes a value to one address or repeatedly across a range.

General syntax:

```
SETMEMORY INITIAL [FINAL] VALUE
```

Parameters:

- INITIAL: first address.
- FINAL: optional last address.
- VALUE: byte sequence in framework numeric syntax; multi-byte values advance by their own length.

Examples:

```
SETMEMORY $0400 $41  
SETMEMORY $0400 $0427 $20
```

Result and consequences: The active memory view is modified immediately; unsafe writes can corrupt state or crash the emulated program.

MEMORYANALYSIS

Availability: Local console and remote /o channel.

Accepted aliases: CMEMORYANALYSIS.

Meaning: Analyzes the current memory configuration.

General syntax:

```
MEMORYANALYSIS
```

Parameters:

- No parameters.

Examples:

```
MEMORYANALYSIS
```

Result and consequences: Analyzes the current memory configuration.

STOP

Availability: Local console and remote /o channel.

Accepted aliases: CSTOP.

Meaning: Stops CPU execution.

General syntax:

```
STOP
```

Parameters:

- No parameters.

Examples:


```
STOP
```

Result and consequences: Stops CPU execution.

RUN

Availability: Local console and remote /o channel.

Accepted aliases: R, CRUN.

Meaning: Starts or resumes CPU execution; alias R.

General syntax:

```
RUN [ADDRESS]
```

Parameters:

- ADDRESS: optional new program counter before running.

Examples:

```
RUN  
R $080d
```

Result and consequences: The CPU resumes immediately, optionally after changing PC.

SETPC

Availability: Local console and remote /o channel.

Accepted aliases: CSETPC.

Meaning: Sets the program counter without starting execution.

General syntax:

```
SETPC ADDRESS
```

Parameters:

- ADDRESS: new CPU address.

Examples:

```
SETPC $080d
```

Result and consequences: Subsequent execution continues from the new address; an unsafe address can break the program.

NEXT

Availability: Local console and remote /o channel.

Accepted aliases: N, CNEXT.

Meaning: Executes the next instruction; alias N.

General syntax:

```
NEXT
```

Parameters:

- No parameters.

Examples:

```
NEXT  
N
```

Result and consequences: Executes the next instruction; alias N.

NEXTSTACK

Availability: Local console and remote /o channel.

Accepted aliases: NS, CNEXTSTACK.

Meaning: Runs until the stack pointer returns to its initial value; alias NS.

General syntax:

```
NEXTSTACK
```

Parameters:

- No parameters.

Examples:

```
NS
```

Result and consequences: Useful for stepping over a routine, although behavior depends on the routine's stack discipline.

SHOWNEXT

Availability: Local console and remote /o channel.

Accepted aliases: SN, CSHOWNEXT.

Meaning: Disassembles upcoming instruction(s); alias SN.

General syntax:

```
SHOWNEXT [NUMBER]
```

Parameters:

- NUMBER: optional count; one instruction by default.

Examples:

```
SHOWNEXT  
SN 8
```

Result and consequences: Disassembles upcoming instruction(s); alias SN.

INST

Availability: Local console and remote /o channel.

Accepted aliases: CINST.

Meaning: Shows the last instruction executed.

General syntax:

```
INST
```

Parameters:

- No parameters.

Examples:

```
INST
```

Result and consequences: Shows the last instruction executed.

BREAKS

Availability: Local console and remote /o channel.

Accepted aliases: CBREAKS.

Meaning: Lists all stop-point addresses.

General syntax:

```
BREAKS
```

Parameters:

- No parameters.

Examples:

```
BREAKS
```

Result and consequences: Lists all stop-point addresses.

SETBREAK

Availability: Local console and remote /o channel.

Accepted aliases: CSETBREAK.

Meaning: Adds one or more breakpoints.

General syntax:

```
SETBREAK ADDRESS1 [ADDRESS2 ...]
```

Parameters:

- ADDRESSn: one or more addresses in framework numeric syntax.

Examples:

```
SETBREAK $080d $ffd2
```

Result and consequences: Adds one or more breakpoints.

REMOVEBREAK

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEBREAK.

Meaning: Removes selected breakpoints.

General syntax:

```
REMOVEBREAK ADDRESS1 [ADDRESS2 ...]
```

Parameters:

- ADDRESSn: one or more existing breakpoint addresses.

Examples:

```
REMOVEBREAK $ffd2
```

Result and consequences: Removes selected breakpoints.

REMOVEBREAKS

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEBREAKS.

Meaning: Removes every breakpoint.

General syntax:

```
REMOVEBREAKS
```

Parameters:

- No parameters.

Examples:

```
REMOVEBREAKS
```

Result and consequences: Removes every breakpoint.

SPEED

Availability: Local console and remote /o channel.

Accepted aliases: CSPEED.

Meaning: Shows measured CPU and screen-I/O execution speed.

General syntax:

```
SPEED
```

Parameters:

- No parameters.

Examples:

```
SPEED
```

Result and consequences: Shows measured CPU and screen-I/O execution speed.

LOADBIN

Availability: Local console and remote /o channel.

Accepted aliases: CLOADBIN.

Meaning: Loads a raw binary file at an explicit address.

General syntax:

```
LOADBIN FILE ADDRESS
```

Parameters:

- FILE: raw binary path.
- ADDRESS: first destination address.

Examples:

```
LOADBIN charset.bin $2000
```

Result and consequences: Bytes are copied into memory without an embedded load-address header.

SAVEBIN

Availability: Local console and remote /o channel.

Accepted aliases: CSAVEBIN.

Meaning: Saves a memory range as raw binary.

General syntax:

```
SAVEBIN FILE SIZE ADDRESS
```

Parameters:

- FILE: output path.
- SIZE: byte count.
- ADDRESS: first source address.

Examples:

```
SAVEBIN screen.bin 1000 $0400
```

Result and consequences: Saves a memory range as raw binary.

DEBUGON

Availability: Local console and remote /o channel.

Accepted aliases: CACTIVATEDEEPDEBUG.

Meaning: Enables deep execution debugging.

General syntax:

```
DEBUGON FILE CPU [APPEND] [FILTERS...]
```

Parameters:

- FILE: output log.
- CPU: YES includes CPU execution; NO omits it.
- APPEND: optional YES appends, NO overwrites (default).
- D:START-END or ALLDIR: address filter.
- C:IDs or ALLCHIP: chip filter.
- I:IDs or ALLIO: device filter.
- M:IDs or ALLMEM: memory filter.

Examples:

```
DEBUGON msx.log YES NO D:$0800-$0fff ALLCHIP ALLIO ALLMEM
```

Result and consequences: Tracing begins with the selected sources and can substantially slow emulation. It cannot coexist with interrupt-only debugging.

DEBUGOFF

Availability: Local console and remote /o channel.

Accepted aliases: CDESACTIVATEDEEPDEBUG.

Meaning: Stops deep execution debugging.

General syntax:

```
DEBUGOFF
```

Parameters:

- No parameters.

Examples:

```
DEBUGOFF
```

Result and consequences: Stops deep execution debugging.

IDEBUGON

Availability: Local console and remote /o channel.

Accepted aliases: CIDEBUGON.

Meaning: Enables interrupt-only debugging.

General syntax:

```
IDEBUGON FILE [APPEND]
```

Parameters:

- FILE: output log.
- APPEND: optional YES appends; NO overwrites (default).

Examples:

```
IDEBUGON interrupts.log NO
```

Result and consequences: Interrupt tracing begins and cannot coexist with deep debugging.

IDEBUGOFF

Availability: Local console and remote /o channel.

Accepted aliases: CIDEBUGOFF.

Meaning: Stops interrupt-only debugging.

General syntax:

```
IDEBUGOFF
```

Parameters:

- No parameters.

Examples:

```
IDEBUGOFF
```

Result and consequences: Stops interrupt-only debugging.

RESTART

Availability: Local console and remote /o channel.

Accepted aliases: CRESTART.

Meaning: Requests a restart and exits the current computer loop so it can be rebuilt.

General syntax:

```
RESTART STOPPED [LEVEL]
```

Parameters:

- STOPPED: YES restarts stopped; NO restarts running.
- LEVEL: optional 0 CPU-only/mild restart (default), 1 computer restart preserving memory, greater than 1 full restart.

Examples:

```
RESTART YES 0  
RESTART NO 2
```

Result and consequences: Requests a restart and exits the current computer loop so it can be rebuilt.

DEVICES

Availability: Local console and remote /o channel.

Accepted aliases: CDEVICES.

Meaning: Shows all or selected I/O devices.

General syntax:

```
DEVICES [ID1 ID2 ...]
```

Parameters:

- IDn: optional device identifiers; with none, all devices are returned.

Examples:

```
DEVICES  
DEVICES 1 2
```

Result and consequences: Shows all or selected I/O devices.

PERIPHERALS

Availability: Local console and remote /o channel.

Accepted aliases: CPERIPHERALS.

Meaning: Shows all or selected connected peripherals.

General syntax:

```
PERIPHERALS [ID1 ID2 ...]
```

Parameters:

- IDn: optional peripheral identifiers; with none, all are returned.

Examples:

```
PERIPHERALS  
PERIPHERALS 100
```

Result and consequences: Shows all or selected connected peripherals.

PERCOMMAND

Availability: Local console and remote /o channel.

Accepted aliases: CPERCMD.

Meaning: Sends a device-specific command to a connected peripheral.

General syntax:

```
PERCOMMAND PERID CMDID [ATTRS...]
```

Parameters:

- PERID: connected peripheral ID.
- CMDID: command ID implemented by that peripheral.
- ATTRS: optional positional attributes forwarded unchanged.

Examples:

```
PERCOMMAND 100 0
```

Result and consequences: The peripheral executes the request if both IDs and its parameter contract are valid; otherwise the answer reports an error.

ASSIGNJOY

Availability: Local console and remote /o channel.

Accepted aliases: CASSIGNJ.

Meaning: Remaps a joystick identifier.

General syntax:

```
ASSIGNJOY ID NEWID
```

Parameters:

- ID: current joystick ID.
- NEWID: required target ID; using the same ID removes an existing assignment.

Examples:

```
ASSIGNJOY 0 1
```

Result and consequences: Remaps a joystick identifier.

CLOCKFACTOR

Availability: Local console and remote /o channel.

Accepted aliases: CCLOCKFACTOR.

Meaning: Changes the CPU clock multiplier.

General syntax:

```
CLOCKFACTOR VALUE
```

Parameters:

- VALUE: numeric factor; 1 means normal speed.

Examples:

```
CLOCKFACTOR 2
```


Result and consequences: The emulation pacing target changes; values above one request faster execution.

SOUNDON

Availability: Local console and remote /o channel.

Accepted aliases: CSOUNDON.

Meaning: Enables MSX PSG sound output.

General syntax:

```
SOUNDON
```

Parameters:

- No parameters.

Examples:

```
SOUNDON
```

Result and consequences: Enables the AY-3-8910 sound device.

SOUNDOFF

Availability: Local console and remote /o channel.

Accepted aliases: CSOUNDOFF.

Meaning: Disables MSX PSG sound output.

General syntax:

```
SOUNDOFF
```

Parameters:

- No parameters.

Examples:

```
SOUNDOFF
```

Result and consequences: Disables audible PSG output while emulation continues.

INTERRUPTS

Availability: Local console and remote /o channel.

Accepted aliases: CINTERRUPTS.

Meaning: Shows all interrupts known by the computer.

General syntax:

```
INTERRUPTS
```

Parameters:

- No parameters.

Examples:

```
INTERRUPTS
```

Result and consequences: Shows all interrupts known by the computer.

INTSET

Availability: Local console and remote /o channel.

Accepted aliases: CINTSET.

Meaning: Enables or disables one interrupt or every interrupt.

General syntax:

```
INTSET ID|ALL ON|OFF
```

Parameters:

- ID or ALL: target interrupt.
- ON or OFF: requested state.

Examples:

```
INTSET ALL ON
```

Result and consequences: An unknown interrupt ID produces an error.

CHIPS

Availability: Local console and remote /o channel.

Accepted aliases: CCHIPS.

Meaning: Lists the emulated chips and their identifying attributes.

General syntax:

```
CHIPS
```

Parameters:

- No parameters.

Examples:

```
CHIPS
```

Result and consequences: Lists the emulated chips and their identifying attributes.

CRTON

Availability: Local console and remote /o channel.

Accepted aliases: CCRTON.

Meaning: Enables the CRT screen effect.

General syntax:

```
CRTON
```

Parameters:

- No parameters.

Examples:

```
CRTON
```

Result and consequences: Enables the CRT screen effect.

CRTOFF

Availability: Local console and remote /o channel.

Accepted aliases: CCRTOFF.

Meaning: Disables the CRT screen effect.

General syntax:

```
CRTOFF
```

Parameters:

- No parameters.

Examples:

```
CRTOFF
```

Result and consequences: Disables the CRT screen effect.

DATASETTE

Availability: Local console and remote /o channel.

Accepted aliases: CDATASETTE.

Meaning: Shows datasette state when a supported datasette is connected.

General syntax:

```
DATASETTE
```

Parameters:

- No parameters.

Examples:

```
DATASETTE
```

Result and consequences: Shows datasette state when a supported datasette is connected.

GRIDON

Availability: Local console and remote /o channel.

Accepted aliases: CGRIDON.

Meaning: Enables the diagnostic grid overlay on the screen.

General syntax:

```
GRIDON COLOR
```

Parameters:

- COLOR: mandatory numeric grid color.

Examples:

```
GRIDON 14
```

Result and consequences: Enables the standard screen grid with the requested color.

GRIDOFF

Availability: Local console and remote /o channel.

Accepted aliases: CGRIDOFF.

Meaning: Disables the diagnostic grid overlay.

General syntax:

```
GRIDOFF
```

Parameters:

- No parameters.

Examples:

```
GRIDOFF
```

Result and consequences: Disables the standard screen grid.

PICTURE

Availability: Local console and remote /o channel.

Accepted aliases: CPICTURE.

Meaning: Saves a screenshot of the emulator screen.

General syntax:

```
PICTURE FILE
```

Parameters:

- FILE: output PNG, JPG or BMP filename.

Examples:

```
PICTURE msx-screen.png
```

Result and consequences: A supported extension writes the current screen; unsupported formats fail.

SETHOOK

Availability: Local console and remote /o channel.

Accepted aliases: CSETHOOK.

Meaning: Creates an execution hook.

General syntax:

```
SETHOOK ID TYPE [PARAMS...]
```

Parameters:

- ID: unique hook identifier.
- TYPE: hook kind; currently 0 stops on memory read/write.
- For TYPE 0, PARAMS are mandatory INITIAL and optional FINAL addresses.

Examples:

```
SETHOOK 1 0 $d000 $d3ff
```

Result and consequences: Creates an execution hook.

REMOVEHOOK

Availability: Local console and remote /o channel.

Accepted aliases: CREMOVEHOOK.

Meaning: Removes a hook by identifier.

General syntax:

```
REMOVEHOOK ID
```

Parameters:

- ID: hook identifier.

Examples:

```
REMOVEHOOK 1
```

Result and consequences: If the hook does not exist, no state changes.

HOOKS

Availability: Local console and remote /o channel.

Accepted aliases: CHOOKS.

Meaning: Lists active hooks.

General syntax:

```
HOOKS
```

Parameters:

- No parameters.

Examples:

```
HOOKS
```

Result and consequences: Lists active hooks.

HOOKSHELP

Availability: Local console and remote /o channel.

Accepted aliases: CHOOKSHELP.

Meaning: Shows help for supported hook types.

General syntax:

```
HOOKSHELP
```

Parameters:

- No parameters.

Examples:

```
HOOKSHELP
```

Result and consequences: Shows help for supported hook types.

CMOVEPARAMS

Availability: Local console and remote /o channel.

Accepted aliases: No public alias.

Meaning: Builds the framework's internal MoveParametersToAnswerCommand.

General syntax:

```
CMOVEPARAMS
```

Parameters:

- No parameters are executable under the current command contract.

Examples:

```
CMOVEPARAMS
```

Result and consequences: The command returns an empty answer when invoked directly; it is an internal response helper and is not advertised in the help file.

3.2 MSX-Specific Commands

VDP

Availability: Local console and remote /o channel.

Accepted aliases: CVDP.

Meaning: Returns the current state of the MSX video display processor.

General syntax:

```
VDP
```

Parameters:

- No parameters.

Examples:

```
VDP
```

Result and consequences: Returns the active TMS9918A/TMS9929A VDP information structure, including registers and current video state.

PSG

Availability: Local console and remote /o channel.

Accepted aliases: CPSG.

Meaning: Returns the current state of the MSX programmable sound generator.

General syntax:

```
PSG
```

Parameters:

- No parameters.

Examples:

PSG

Result and consequences: Returns the AY-3-8910 PSG information structure and its current register-derived state.

PNAME

Availability: Local console and remote /o channel.

Accepted aliases: CPNAME.

Meaning: Returns a snapshot of the VDP pattern-name table selected by the current VDP registers.

General syntax:

PNAME

Parameters:

- No parameters.

Examples:

PNAME

Result and consequences: Returns the bytes in the active pattern-name table. Their interpretation depends on the current VDP screen mode.

PGEN

Availability: Local console and remote /o channel.

Accepted aliases: CPGEN.

Meaning: Returns a snapshot of the VDP pattern-generator table selected by the current VDP registers.

General syntax:

PGEN

Parameters:

- No parameters.

Examples:

PGEN

Result and consequences: Returns the bytes used by the active VDP pattern-generator table.

CNAME

Availability: Local console and remote /o channel.

Accepted aliases: CCNAME.

Meaning: Returns a snapshot of the VDP colour table selected by the current VDP registers.

General syntax:

CNAME

Parameters:

- No parameters.

Examples:

```
CNAME
```

Result and consequences: Returns the bytes in the active colour table. Their interpretation depends on the current VDP screen mode.

SPRITESDRAW

Availability: Local console and remote /o channel.

Accepted aliases: CSPRITESDRAW.

Meaning: Returns textual drawings of all or selected MSX sprite patterns.

General syntax:

```
SPRITESDRAW [1..32 ...]
```

Parameters:

- Sprite numbers: optional values from 1 through 32. Invalid values are ignored and duplicates are coalesced; with none, the command returns all sprites.

Examples:

```
SPRITESDRAW  
SPRITESDRAW 1 8 16 32
```

Result and consequences: Returns a textual representation of the selected sprite patterns using the VDP's current sprite tables and mode.

VDPMEMORY

Availability: Local console and remote /o channel.

Accepted aliases: CVDPMEMORY.

Meaning: Reads a byte or inclusive range from VDP video memory.

General syntax:

```
VDPMEMORY ADDRESS [OTHER_ADDRESS]
```

Parameters:

- ADDRESS: mandatory first video-memory address.
- OTHER_ADDRESS: optional second endpoint. Reversed endpoints are normalized. The parser accepts the Z80's 16-bit address range; meaningful mapped VRAM depends on the active VDP.

Examples:

```
VDPMEMORY 0x0000  
VDPMEMORY 0x1800 0x181F
```

Result and consequences: Returns the addressed VDP-memory bytes; one endpoint reads one byte and two endpoints read the inclusive normalized range.

SETVDPMEMORY

Availability: Local console and remote /o channel.

Accepted aliases: CVDPSETMEMORY.

Meaning: Writes one byte to one VDP-memory address or repeatedly across an inclusive range.

General syntax:

```
SETVDPMEMORY ADDRESS [FINAL_ADDRESS] VALUE
```

Parameters:

- ADDRESS: mandatory first video-memory address.
- FINAL_ADDRESS: optional range endpoint. Reversed endpoints are normalized.
- VALUE: mandatory byte value written at every selected address.

Examples:

```
SETVDPMEMORY 0x1800 0x41
SETVDPMEMORY 0x1800 0x181F 0
```

Result and consequences: Changes the selected VRAM byte or fills the inclusive range; malformed parameters return an error response.

VDPEVENTS

Availability: Local console and remote /o channel.

Accepted aliases: CVDPEVENTS.

Meaning: Enables or disables visualization of significant CPU/VDP events.

General syntax:

```
VDPEVENTS ON|OFF
```

Parameters:

- ON enables event visualization. Any other single token, including OFF, disables it.

Examples:

```
VDPEVENTS ON
VDPEVENTS OFF
```

Result and consequences: Changes the event-overlay flag used by the active MSX VDP.

3.3 Local-Console-Only Commands

LOADPRG

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads, assembles and installs an assembler source through the emulator object.

General syntax:

```
LOADPRG FILE
```

Parameters:

- FILE: assembler source path.

Examples:

```
LOADPRG demo.asm
```

Result and consequences: Compilation or loading errors are reported by the local console.

LOADBINARY

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads a binary containing its destination address in the file header.

General syntax:

```
LOADBINARY FILE
```

Parameters:

- FILE: header-addressed binary path.

Examples:

```
LOADBINARY demo.bin
```

Result and consequences: Loads a binary containing its destination address in the file header.

LOADBLOCKS

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads a block-structured memory file.

General syntax:

```
LOADBLOCKS FILE
```

Parameters:

- FILE: block file path.

Examples:

```
LOADBLOCKS state.blocks
```

Result and consequences: Loads a block-structured memory file.

DECOMPILE

Availability: Local console only.

Accepted aliases: None.

Meaning: Disassembles a memory region.

General syntax:

```
DECOMPILE ADDRESS BYTES
```

Parameters:

- ADDRESS: first address.
- BYTES: approximate byte count; output rounds to complete instructions.

Examples:

```
DECOMPILE $080d 64
```

Result and consequences: Disassembles a memory region.

POSSIBLEPERS

Availability: Local console only.

Accepted aliases: None.

Meaning: Lists the peripheral IDs advertised by the MSX I/O peripheral builder.

General syntax:

```
POSSIBLEPERS
```

Parameters:

- No parameters.

Examples:

```
POSSIBLEPERS
```

Result and consequences: Lists ID 10 (typewriter keyboard injection) and ID 101 (MSX CAS cassette injection).

CONNECTPER

Availability: Local console only.

Accepted aliases: None.

Meaning: Builds and connects an MSX peripheral to the first compatible device.

General syntax:

```
CONNECTPER ID [ATTRS...]
```

Parameters:

- ID: 10 for typewriter input or 101 for CAS cassette injection.
- ATTRS: optional construction attributes; neither currently advertised peripheral requires one.

Examples:

```
CONNECTPER 10  
CONNECTPER 101
```

Result and consequences: Connection succeeds only when the ID exists and a compatible target device is available.

DISCONNECTPERS

Availability: Local console only.

Accepted aliases: None.

Meaning: Disconnects one or more peripherals.

General syntax:

```
DISCONNECTPERS ID1 [ID2 ...]
```

Parameters:

- IDn: connected peripheral IDs.

Examples:

```
DISCONNECTPERS 100 101
```

Result and consequences: Disconnects one or more peripherals.

LOADPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Loads keystrokes or cassette media into a connected MSX peripheral.

General syntax:

```
LOADPERDATA ID FILE [DEBUGFILE]
```

Parameters:

- ID: connected peripheral ID.
- FILE: a supported keystroke file for ID 10 or an MSX CAS file for ID 101.
- DEBUGFILE: optional trace file enabled after a successful load.

Examples:

```
LOADPERDATA 10 startup.txt  
LOADPERDATA 101 program.cas cassette-trace.log
```

Result and consequences: Loads the file only if the selected peripheral accepts its type. ID 101 handles CAS blocks through the standard MSX cassette BIOS entry points.

EMPTYPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Creates and attaches an empty CAS data container to a connected cassette-injection peripheral.

General syntax:

```
EMPTYPERDATA ID FILE
```

Parameters:

- ID: normally 101.
- FILE: destination path; use the .cas extension.

Examples:

```
EMPTYPERDATA 101 recording.cas
```

Result and consequences: Prepares empty writable cassette data. The emulated BIOS TAPOON/TAPOUT/TAPOOF calls can then record blocks into it.

SAVEPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Saves the current data retrieved from a connected peripheral.

General syntax:

```
SAVEPERDATA ID [FILE]
```

Parameters:

- ID: connected peripheral ID, normally 101 for cassette data.
- FILE: optional output path; it may be omitted after LOADPERDATA or EMPTYPERDATA established a filename.

Examples:

```
SAVEPERDATA 101  
SAVEPERDATA 101 copy.cas
```

Result and consequences: Writes the cassette's current CAS blocks when the format is writable; otherwise an error is returned.

CLEARPERDATA

Availability: Local console only.

Accepted aliases: None.

Meaning: Clears all data currently loaded into peripherals.

General syntax:

```
CLEARPERDATA
```

Parameters:

- No parameters.

Examples:

```
CLEARPERDATA
```

Result and consequences: Commonly used to force a subsequent program/media reload.

Appendices

Appendix A. Command Availability Summary

The table distinguishes the interactive console from the command channel of the non-console executable. Remote availability assumes MSXEmulator.exe was started with /o and that the client is connected.

Command family	MSXEmulatorC local console	MSXEmulator remote /o
Standard builder	Yes	Yes
MSX builder	Yes	Yes
LocalConsole emulator/media operations	Yes	No

Appendix B. Source-of-Truth Maintenance Rule

This file is a maintained repository artifact. Every change that adds, removes, renames or changes an MSX startup option, model selector, command-builder registration, LocalConsole-only command, syntax, parameter meaning, consequence or visible formatted output must include a review and, when applicable, an update of docs/MSXData/MSXEmulator_UserGuide.docx. Update the corresponding canonical .fmt source whenever the command's InfoStructure contract changes.

Audit the complete chain, not only the edited class: both MSXEmulator entry points and the MSX::MSXEmulator hierarchy for startup options; LocalConsole for local-only commands; and the complete effective MSX::CommandBuilder -> MCHEmul::StandardCommandBuilder -> MCHEmul::StandardCommandBuilder chain for builder commands.

Illustrations

NO ILLUSTRATIONS ARE INCLUDED IN THIS EDITION.

IGNACIO CEA FORNIÉS

Equations

NO EQUATIONS ARE INCLUDED IN THIS EDITION.

IGNACIO CEA FORNIÉS